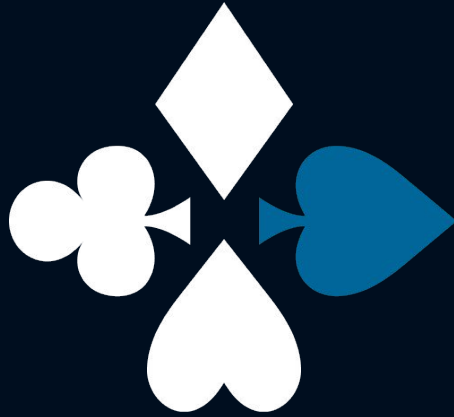




# Pokerbots 2026

Lecture 8: GTO Wizard

# Sponsors



CITADEL | CITADEL Securities



# Announcements

# Lightning Tournament #1

|                                    |       |
|------------------------------------|-------|
| 1. hey0999                         | \$400 |
| 2. Policy Gradient Is All You Need | \$300 |
| 3. Chasers 1/2 Regs                | \$225 |
| 4. spongebot                       | \$175 |
| 5. Los Camellos                    | \$150 |

# Week 2 Mini Tournament Results

WINNER [\$1000]:

heyo999

BIGGEST UPSET [\$500]:

hk

Tournament ELO 936, won majority matches against “Suits GPT” (ELO 1941), 30-0!

MOST IMPROVED [\$750]:

hi

Tournament ELO 1523 → 1939

# Hackathon!



You just had  
to be there



Another???

**[pkr.bot/hack](http://pkr.bot/hack)**

# Make-up Progress Reports due Tuesday

- As per syllabus, if your team missed the Week 2 bot submission deadline, a progress report is required to earn credit
- Reports are half a page and must list team name and members
- Submit via Canvas by 11:59pm Tuesday (Tomorrow)



Giveaway Time  
:D



# Guess Your Guess

- Submit an integer
- Winner is whoever guesses closest to their submission 'index' (in the case of ties, the later submission wins)
  - Allowed to guess decimal numbers
  - Distance is just absolute difference
  - i.e. if you are the 15th person to fill out the form and guess 16, your distance is 1
- Prize: Logitech G305 Lightspeed Mouse!

Questions?

[pkr.bot/index](https://pkr.bot/index)



# Today

## Guest Lecture: GTO Wizard

- Current state of the art Poker Solver and Study tool
- Developed by team of top AI researchers and engineers
- Core team will be presenting their journey and research



**10,000,000+ presolved solutions in your pocket.**

Instantly access any spot you can imagine.  
From preflop to any river you want, we have all the possible situations.  
It works seamlessly on your desktop and mobile.

6max Cash

Spins

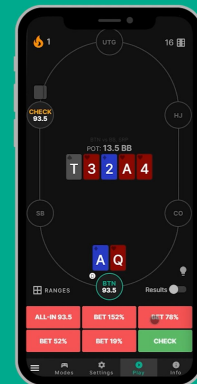
HU SNGs

MTTs

SOON  
Your solutions

PLANNING  
PLO

[Get solutions for free](#)

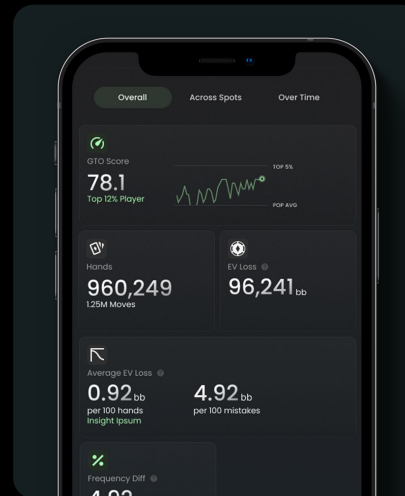
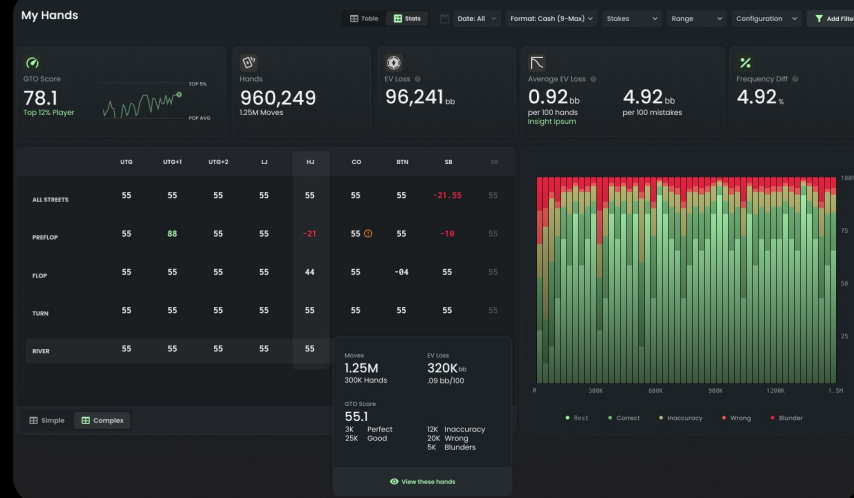


GTO Wizard

# Building a superhuman AI for poker

# About us

- Biggest educational application for poker players
- Founded 5 years ago and currently ~100 employees



No.1 App  
for poker  
players

# About us

- Biggest educational application for poker players
- Founded 5 years ago and currently ~100 employees
- Strong community and partner with some of the biggest players in the industry
- Sponsor major events such as the World Series of Poker and Triton series



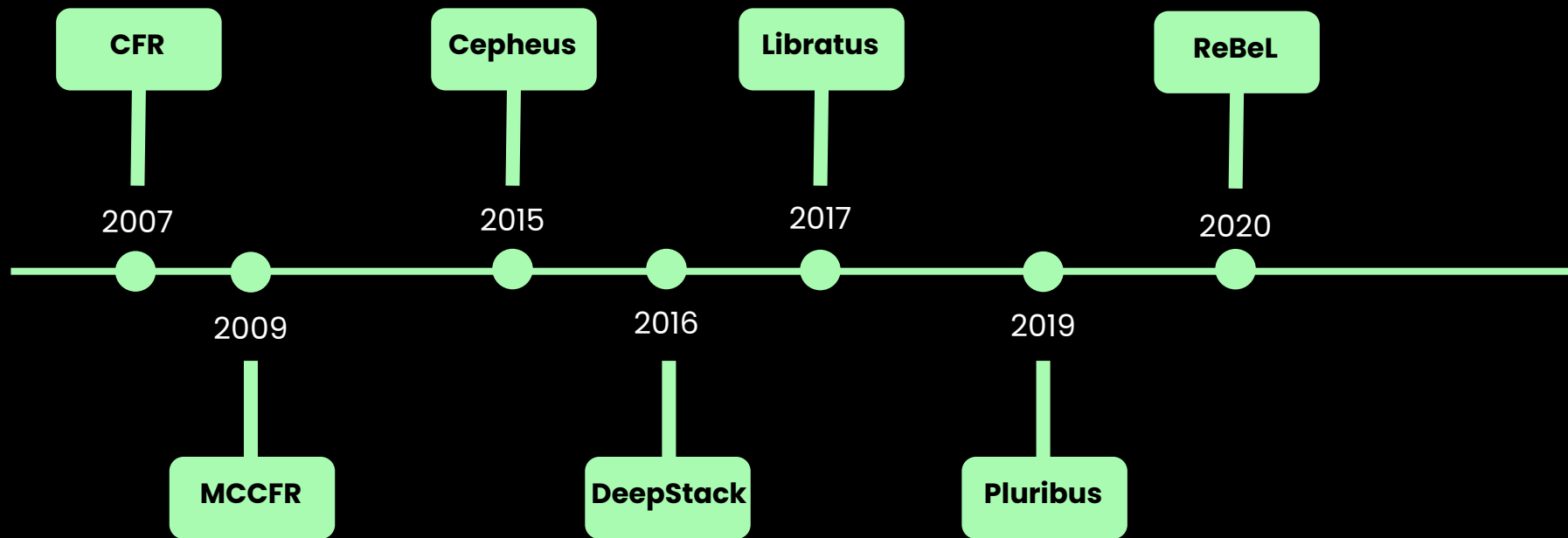
# About us

- About GTO Wizard AI – Team of 11 researchers and engineers
- Built the strongest AI for No Limit Hold'em
- Our goal is to solve any poker variant to a very high accuracy in a few seconds
  - Any number of players
  - Any variant – Hold'em, Omaha, Short-deck, Stud, bomb pots, etc.
  - Any cash game and tournament variations



# Building a superhuman AI for poker

# Timeline of Major Breakthroughs



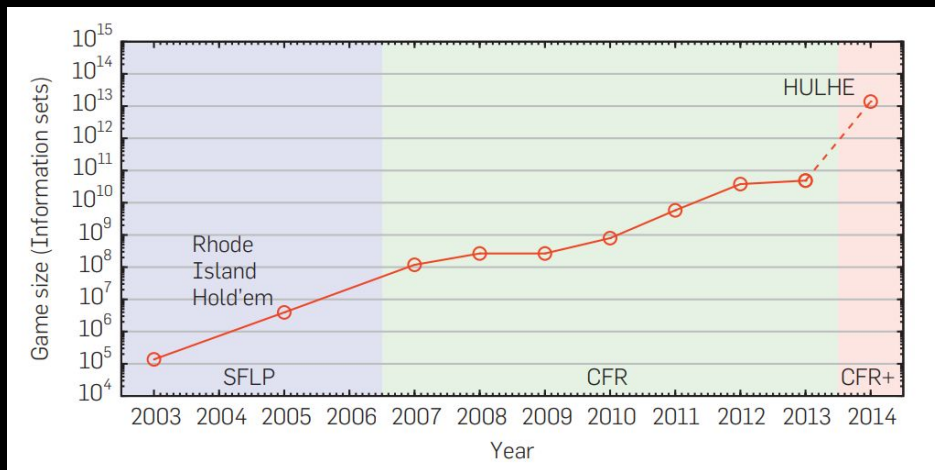


# Cepheus (Bowling et al., 2015)

Essentially solved Heads-Up Limit Hold'em, the smallest variant of poker that humans actually play ( $10^{14}$  decision points).

Thanks to algorithmic advances like CFR+, that allowed to solve games a few orders of magnitude larger.

Storing the strategy would require 262 TiB of memory with 4-byte values.



Size of imperfect-information games that we can exactly solve, plotted over time (Bowling et al., 2015)

→ Problem! No-limit Hold'em has around  $10^{160}$  decision points!

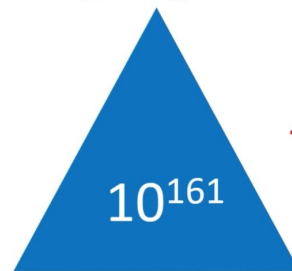
# Solving with Abstractions

Technique used by every winner of the Annual Computer Poker Competition (ACPC) for NLH, until the last year (2018)

- Solve an abstracted version of the game
- At inference time, simply use this as a lookup table

## Abstraction [Gilpin & Sandholm EC-06, *J. of the ACM* 2007...]

Original game



Automated abstraction

Abstracted game



Custom  
equilibrium-finding  
algorithm

~equilibrium

Reverse mapping

$\epsilon$  equilibrium

Foreshadowed by Shi & Littman 01, Billings *et al.* IJCAI-03

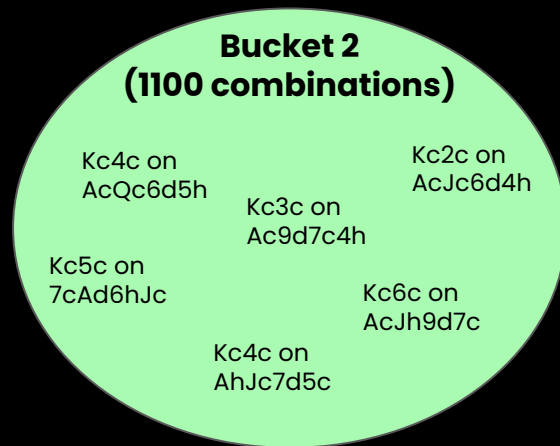
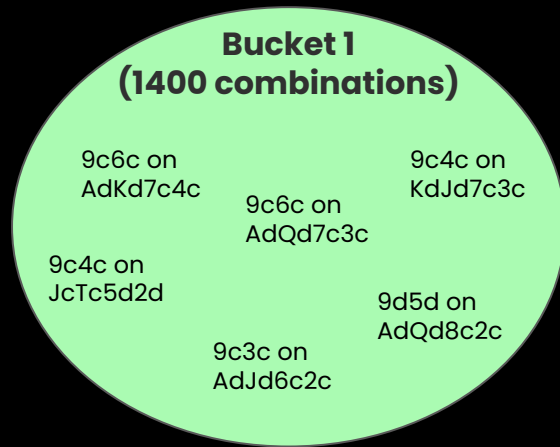
# Information Abstraction

The idea is to group strategically similar decision points

In Hold'em, this represents a mapping from strategically similar {hand, board} combinations to buckets.

When using very large abstractions, these buckets can represent pretty sophisticated categories. Here are two real examples of buckets used by our turn's neural network:

1. 9-high flush draws on a two-broadway board where one board card is lower than the side card
2. King-high flush draws on a Ace-high board where the lowest board card is higher than the side card



# How are these abstractions created?

- Define a set of features for each information set in the original game
  - Define a distance function  $d_{\{i, j\}}$  between pairs of information sets
  - Run a clustering algorithm (e.g. k-means)
- Obtain a many-to-one mapping between the full game's information sets and the information sets in a smaller abstracted game

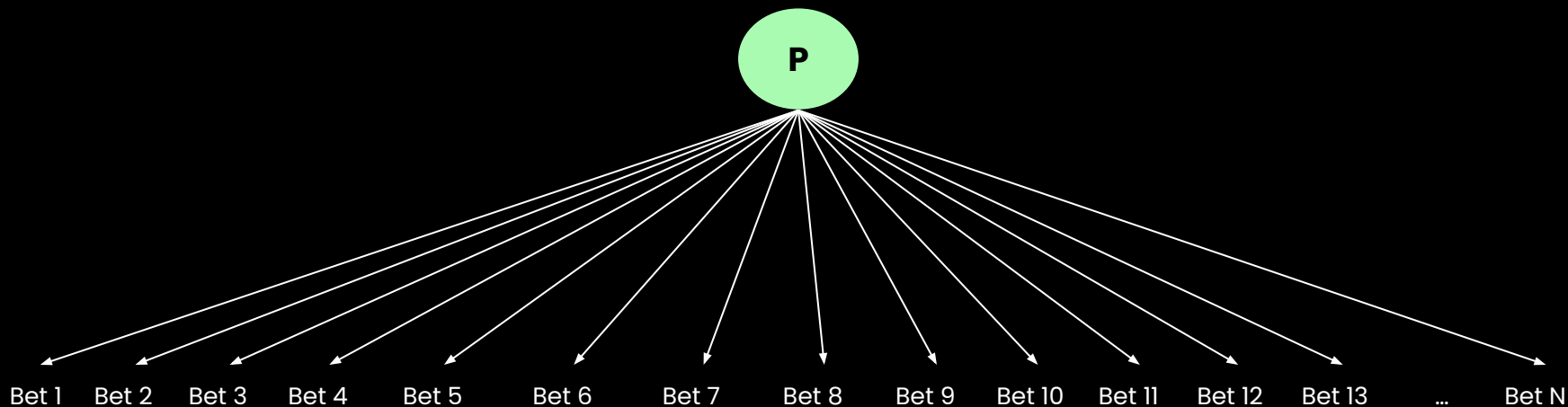
# Example: Expectation-based abstractions

Idea: Group similar states together according to some metric

The most common and obvious one is the **expected hand strength**  $E[HS]$ , commonly called the equity in poker, which is equal to  $P(win) + P(tie)/2$  against a uniform random range.

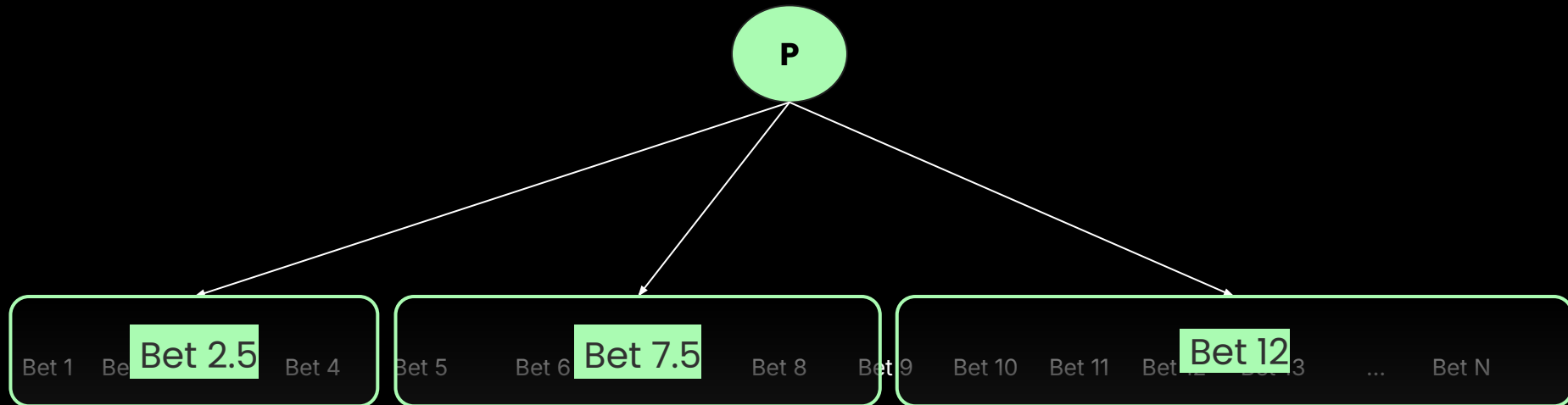
| {Hand, Board}                                                                                  | $E[HS]$ |
|------------------------------------------------------------------------------------------------|---------|
| $\{2\spadesuit 2\clubsuit, K\spadesuit 8\diamondsuit 6\heartsuit 2\heartsuit 5\diamondsuit\}$  | 0.9394  |
| $\{8\spadesuit 8\diamondsuit, J\spadesuit 8\clubsuit 6\heartsuit A\spadesuit T\heartsuit\}$    | 0.9424  |
| $\{J\spadesuit 2\spadesuit, K\spadesuit 8\diamondsuit 6\heartsuit 2\heartsuit 5\diamondsuit\}$ | 0.4192  |
| $\{6\spadesuit 4\spadesuit, J\spadesuit 8\clubsuit 6\heartsuit A\spadesuit T\heartsuit\}$      | 0.4202  |

# Action Abstraction



The action space in no-limit hold'em is huge, but most actions are very strategically similar

# Action Abstraction



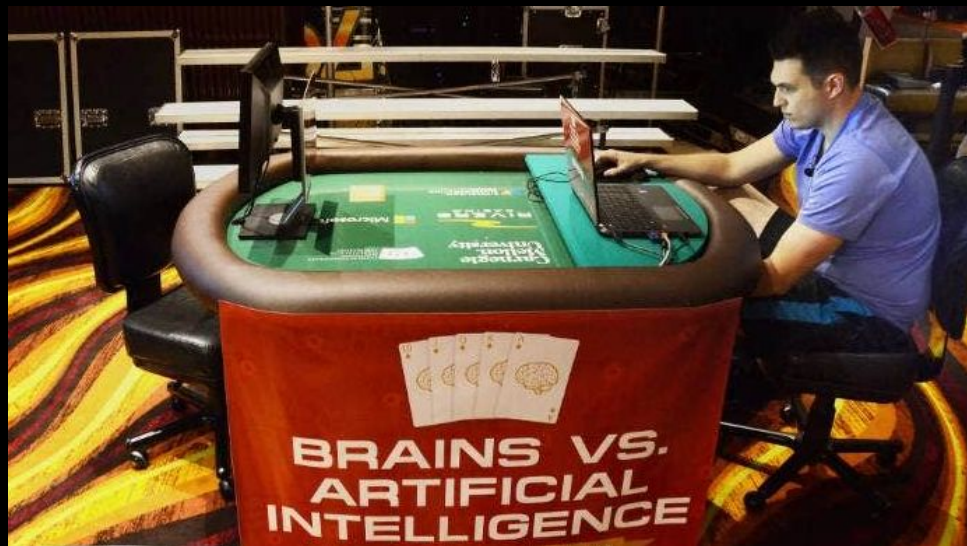
In poker, it's straightforward to group similar actions, but that's not usually the case for other games

# Solving with Abstractions

Not enough to achieve  
superhuman performance

Claudico, a bot from CMU  
leveraging these techniques,  
was tested against four top  
Heads-up NLH professionals.

- Lost by 9 BB/100, a  
significant margin



*Claudico Brain vs AI (2015)*



# Problem with abstraction-based agents

Abstraction-based agents have been shown to be highly exploitable. Against their worst-case opponent, they are worse than the Always Fold strategy!

Local Best Response (LBR) is a technique for finding a lower-bound on a strategy's exploitability

| Agent            | LBR performance (bb/100) |
|------------------|--------------------------|
| Always Fold      | 75.0 $\pm$ 0             |
| Slumbot (2016)   | 402.0 $\pm$ 11.5         |
| Act1 (2016)      | 259.7 $\pm$ 14.0         |
| DeepStack (2017) | -38.3 $\pm$ 21.9         |

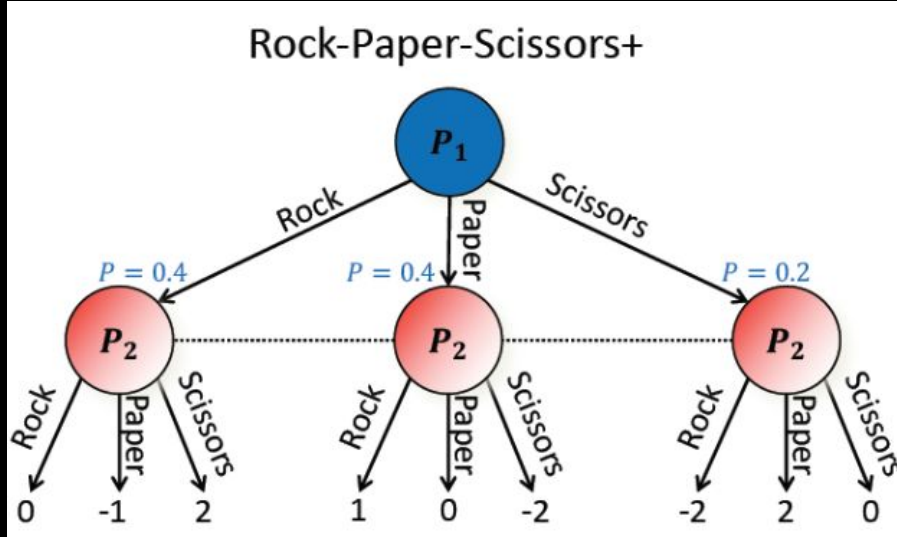
# Why Search is Hard

Search has been used for decades in perfect-information games, so why is it just recent in imperfect-information games?

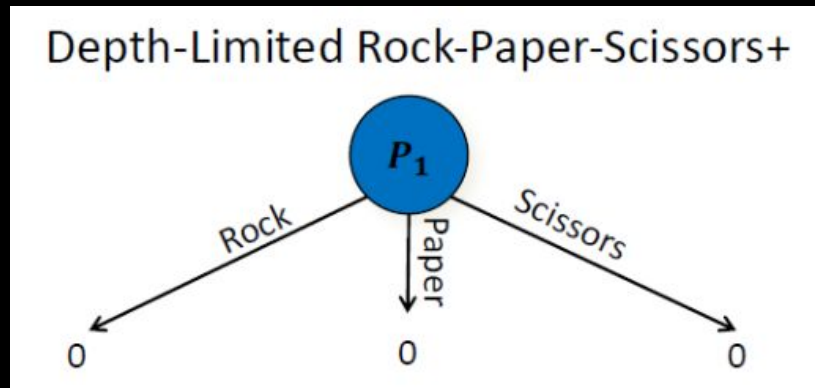
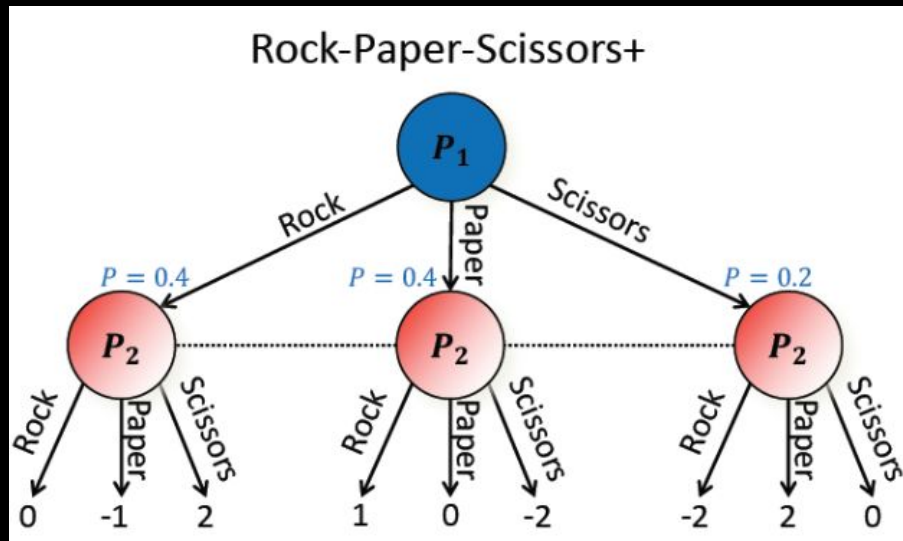
**There are two major problems** in imperfect-information games that make search particularly hard (Sokota et al., 2023):

- **Backward Dependence Problem** (states don't have unique values)
- **Non-Correspondence Problem** (equilibrium selection problem)

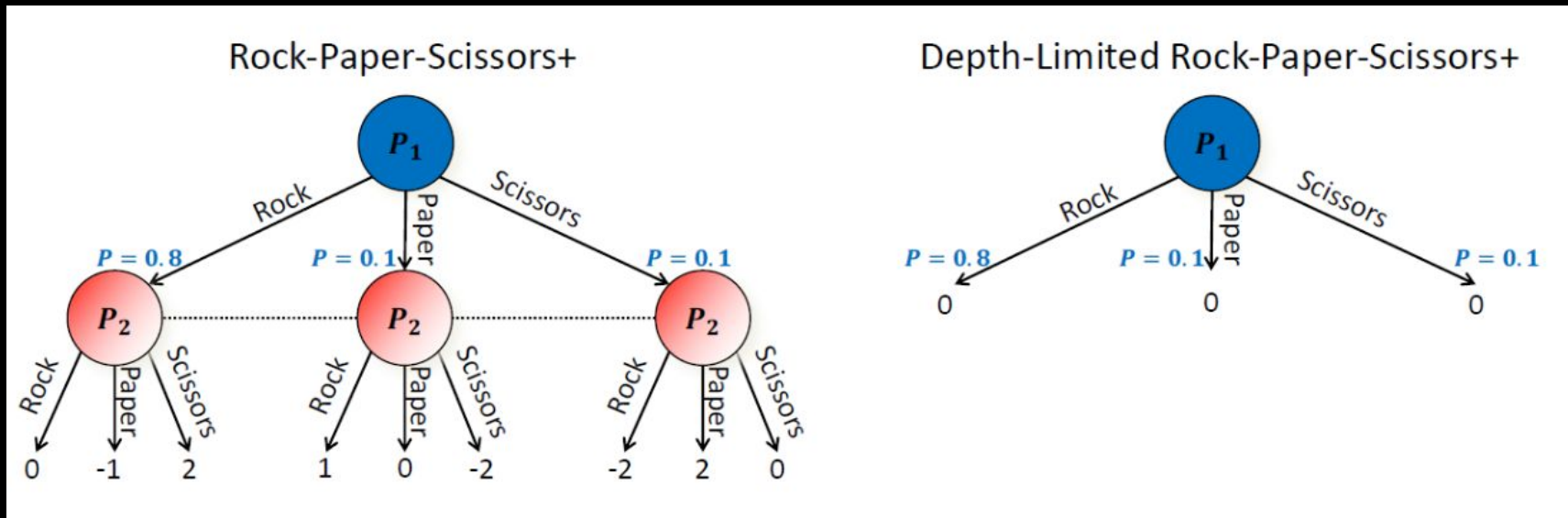
# Depth-Limited Search



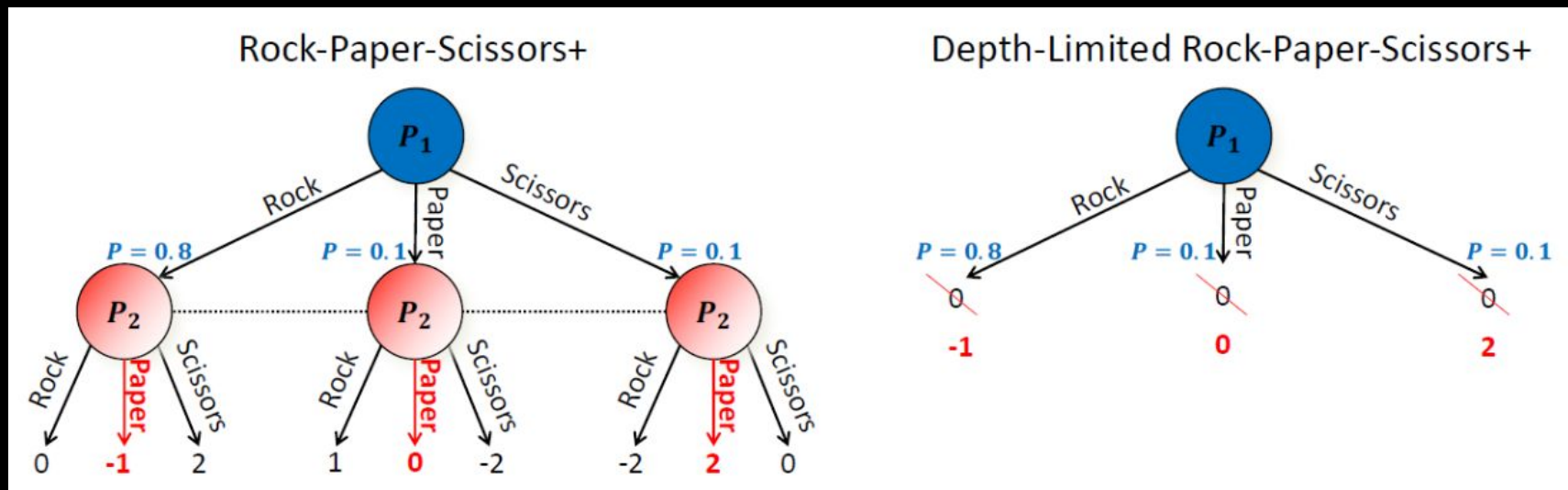
# Depth-Limited Search



# Depth-Limited Search



# Depth-Limited Search



One solution is to condition the value on the probability distribution over states

- $v(\text{Rock})$  is not well defined
- $v(0.8 \text{ Rock}, 0.1 \text{ Paper}, 0.1 \text{ Scissors}) = -0.6$

Taken from (Brown, 2020)

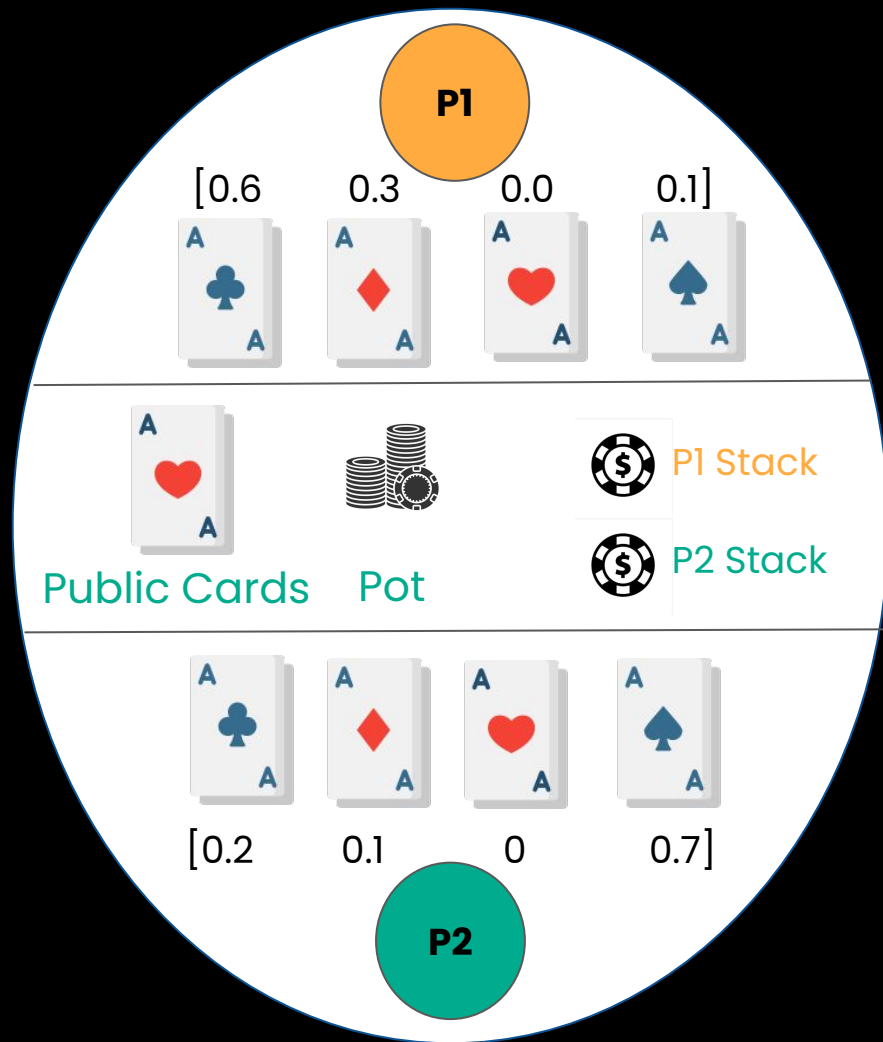
# Public Belief States (PBS)

PBS: A common-knowledge probability distribution over states in some public state.

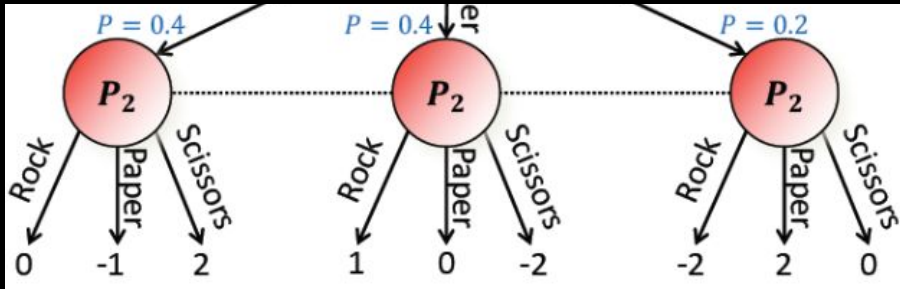
Properties:

- Identical to perfect-info states in perfect-info games
- Have a unique value in two-player zero-sum games

→ Solves the Backward Dependence problem



# Safe Subgame Solving

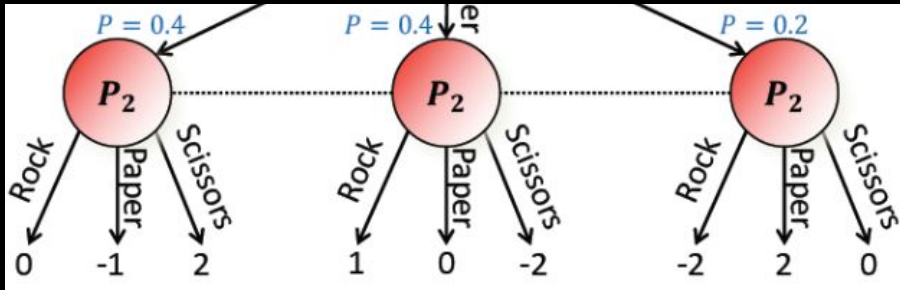


However, let's take the perspective of P2, where we're now resolving the subgame rooted at P2's node. P2 knows that P1 plays Nash  $(0.4, 0.4, 0.2)$ . From P2's POV, in this subgame, any strategy is a Nash equilibrium since all the expected values are equal

- $v(\text{Rock} \mid P1 \text{ plays } [0.4, 0.4, 0.2]) = 0$
- $v(\text{Paper} \mid P1 \text{ plays } [0.4, 0.4, 0.2]) = 0$
- $v(\text{Scissors} \mid P1 \text{ plays } [0.4, 0.4, 0.2]) = 0$



# Safe Subgame Solving



Since any strategy is Nash for  $P_2$ , we need a mechanism for  $P_2$  to select the correct one (0.4, 0.4, 0.2). The issue is that we're assuming  $P_1$  to be fixed but, in reality,  $P_1$  could adapt their strategy in the past to exploit us.

→ This is the **Non-Correspondence Problem**. There exists different techniques attempting to solve it, like augmented subgames (Brown & Sandholm, 2017) and adding regularization objectives to have uniqueness guarantees (Sokota et al., 2023).

# DeepStack (Moravčík et al., 2017)

Algorithm that successfully implement heuristic search in imperfect-information games

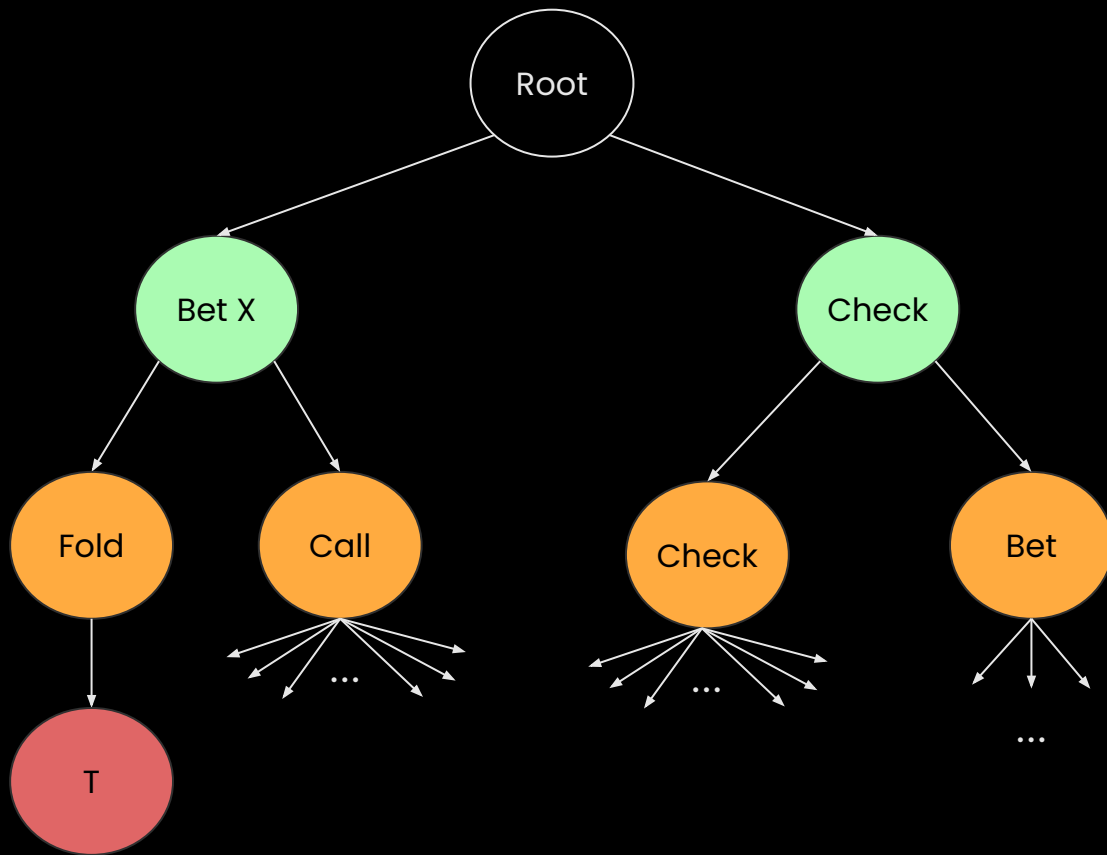
- Introduces a way to decompose the game in depth-limited subgames with CFR-Decomposition.
- Uses a value network to estimate the leaf nodes at a given depth limit.
- The depth-limited subgame is small enough to be solved in real-time “without the need for abstraction”.

# Depth-limited solving with value networks

Player 1

Player 2

Terminal



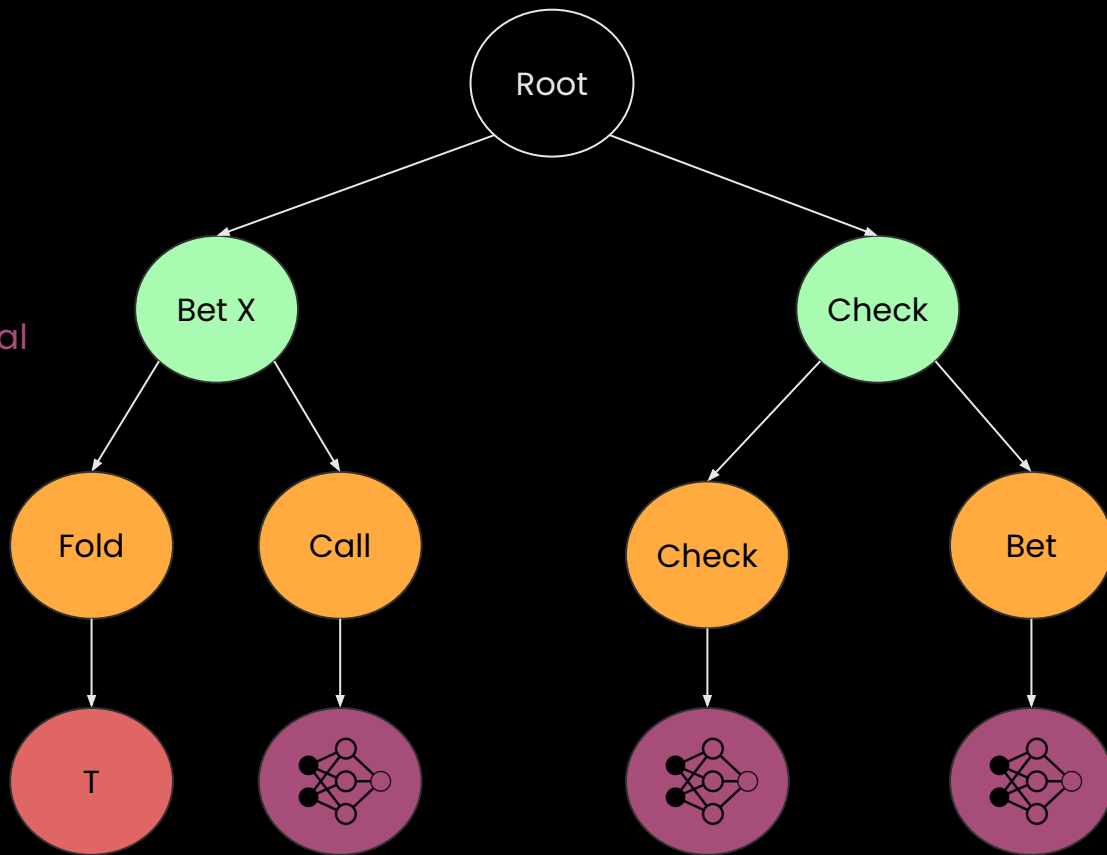
# Depth-limited solving with value networks

Player 1

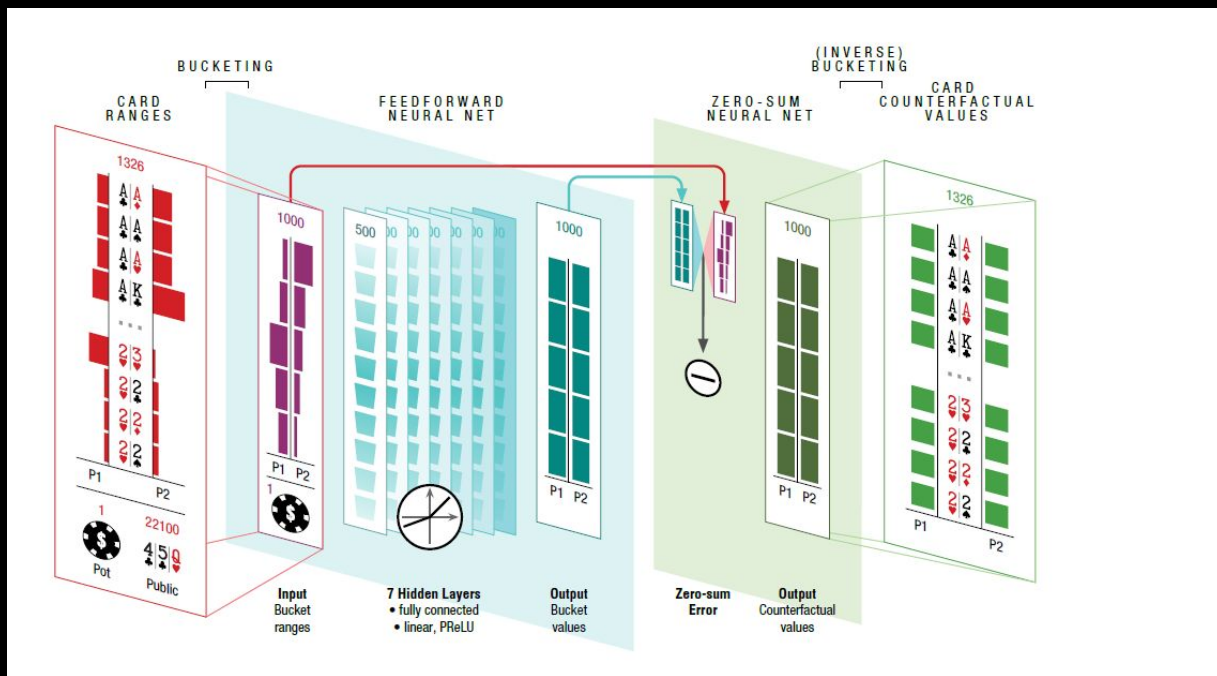
Player 2

Terminal

Pseudo-terminal



# DeepStack's Value Network (Moravčík et al., 2017)



# DeepStack (Moravčík et al., 2017)

While DeepStack showed a general framework to do search in imperfect-information games, its performance was unclear.

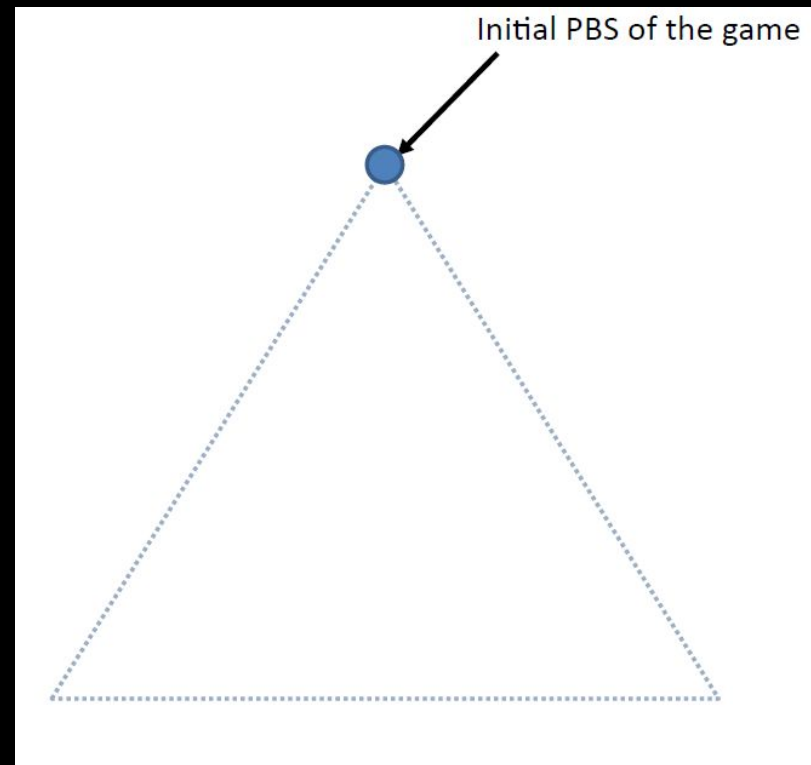
- Evaluated against humans that were not specialized in heads-up poker. Matches were done in an online setting with low financial incentives.
- Not evaluated against other poker bots.
- A reimplementaion of DeepStack actually found it to be losing substantially to Slumbot, the last winner of the ACPC (Zarick & al, 2020)
- Still used a lot of domain knowledge to achieve high performance
- Required multiple steps (far from end-to-end learning)

# ReBeL (Brown et al., 2020)

- General approach for self-play reinforcement learning and search for imperfect-information games
- The algorithm reduces to an algorithm similar to AlphaZero in perfect-information games
- Uses neural networks to predict values at the depth-limit, like DeepStack
- Doesn't use any domain knowledge in the form of information abstraction
- Achieves superhuman performance
- Along with Supremus (Zarick et al., 2020), ReBeL proved that Deep Counterfactual Value Networks can achieve very strong performance, which was doubted by many at the time

# ReBeL (Brown et al., 2020)

Generate a subgame rooted at the start of the game and solve it



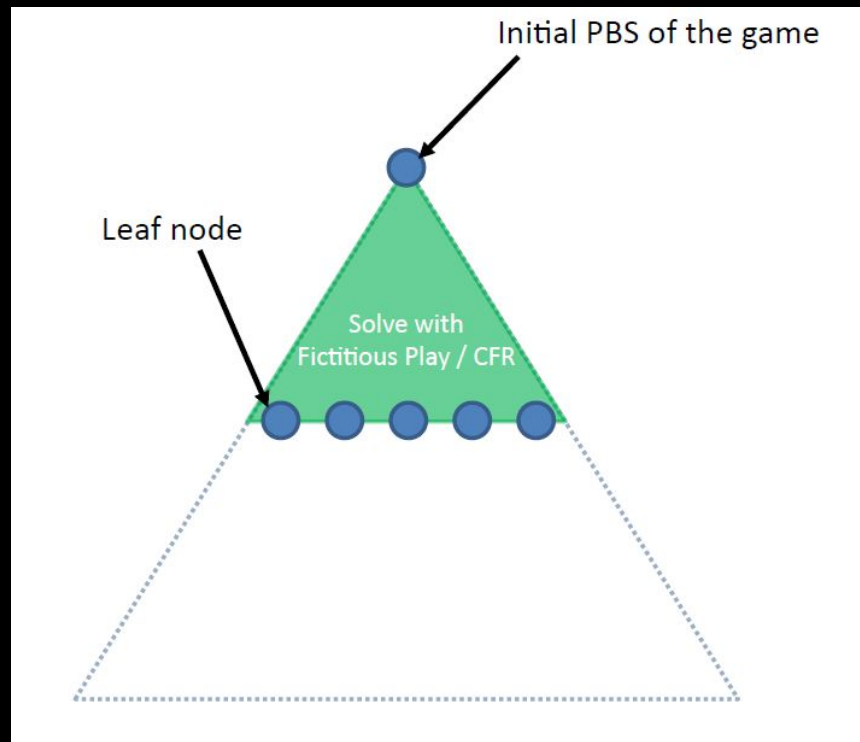
Taken from (Brown, 2020)



# ReBeL (Brown et al., 2020)

Generate a subgame rooted at the start of the game and solve it

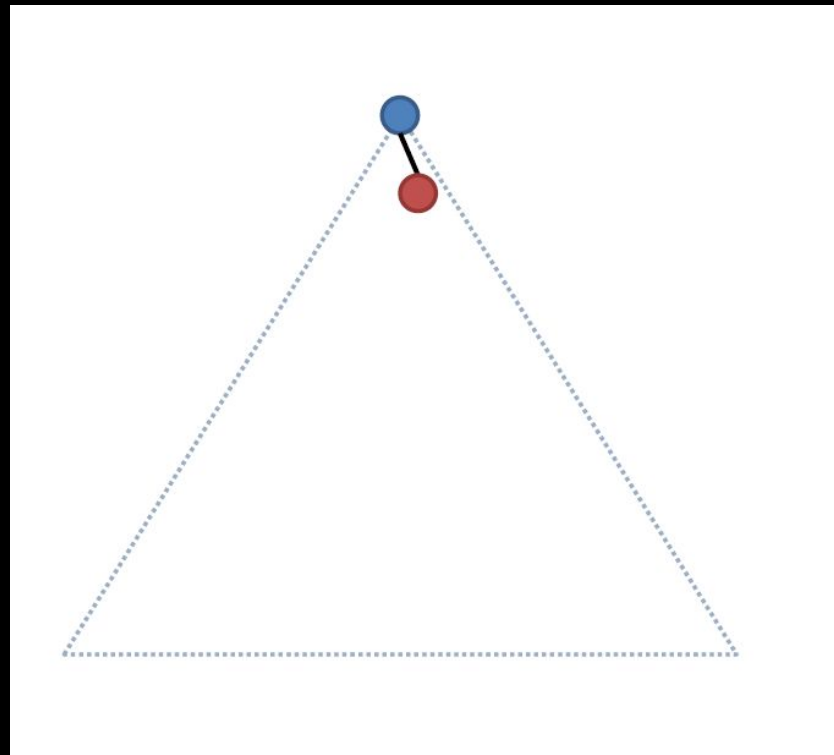
- Solve using CFR
- Leaf values come from value network
- Take next action



# ReBeL (Brown et al., 2020)

After an agent's action, generate a subgame and solve it

- Solve using CFR
- Leaf values come from value network
- Take next action



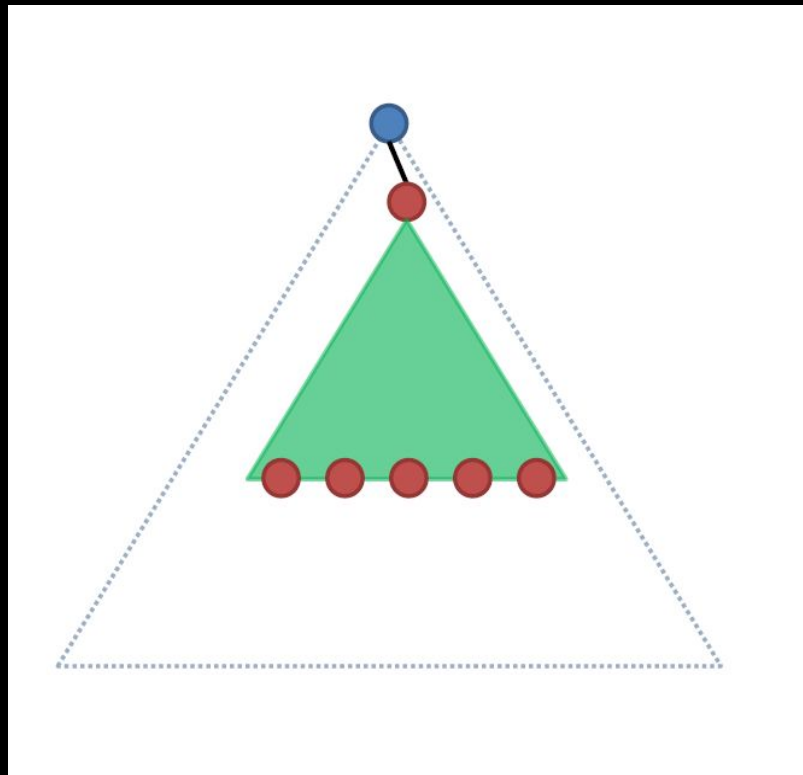
Taken from (Brown, 2020)

# ReBeL (Brown et al., 2020)

After an agent's action, generate a subgame and solve it

- Solve using CFR
- Leaf values come from value network
- Take next action

Repeat until end of the game



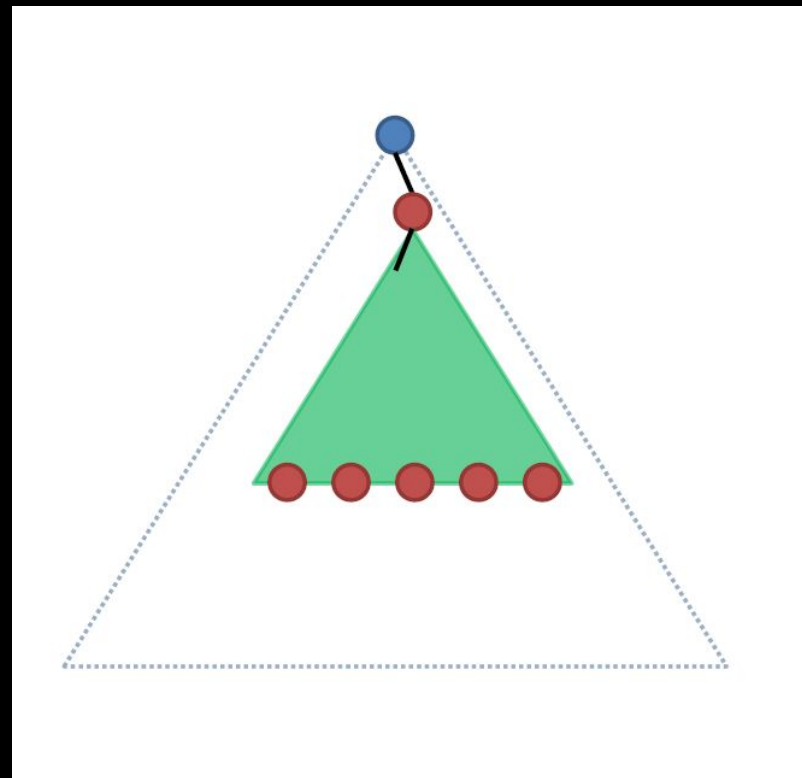
Taken from (Brown, 2020)

# ReBeL (Brown et al., 2020)

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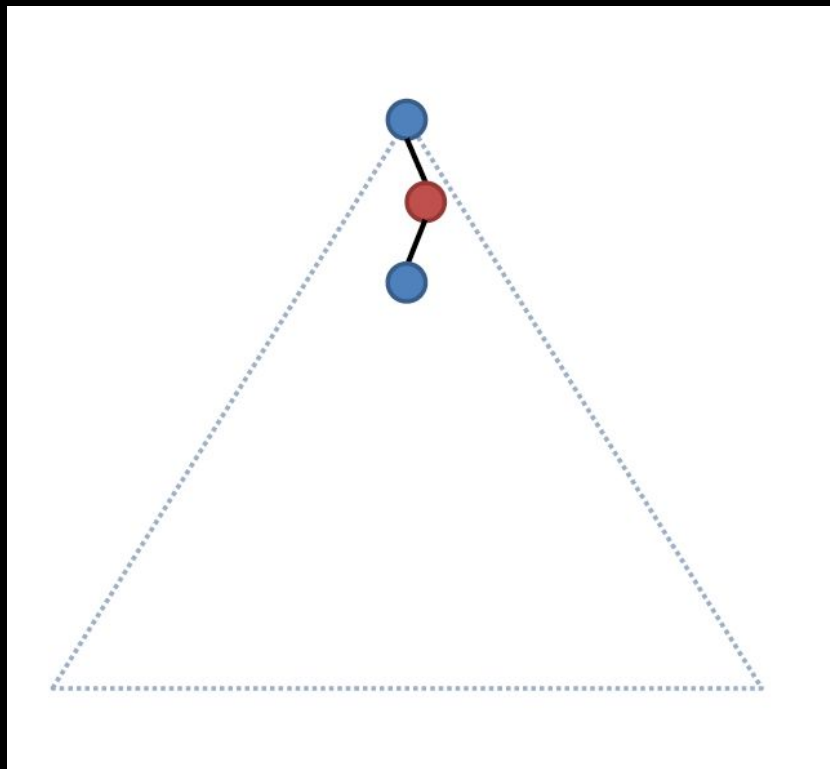
Taken from (Brown, 2020)

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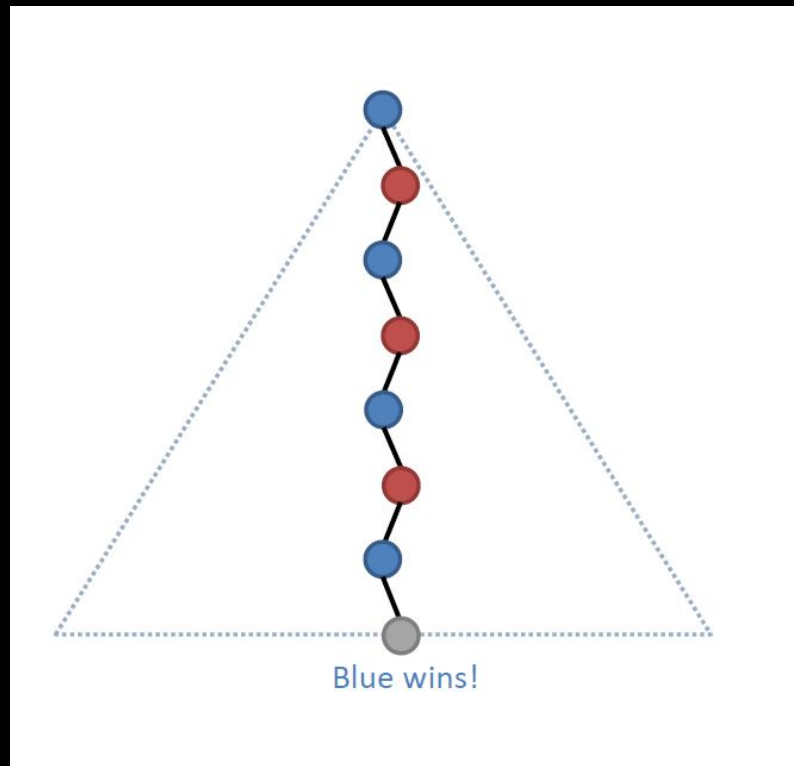
Taken from (Brown, 2020)

# ReBeL (Brown et al., 2020)

After an agent's action, generate a subgame and solve it

- Solve using CFR
- Leaf values come from value network
- Take next action

Final values are used as a training sample for the value network



# GTO Wizard AI

- State-of-the-art hold'em solver, now supporting any number of players (2-9)
- Uses reinforcement learning and search – similarly to ReBeL
- Engineering and algorithmic improvements to achieve state-of-the-art performance with an inference time of a few seconds only, even with 9 players.
- Solves for any standard parameters (rake, antes, straddles, etc.)
- Support for solving under the Independent Chip Model (ICM), which is the most accepted model for translating chips into dollars in tournaments



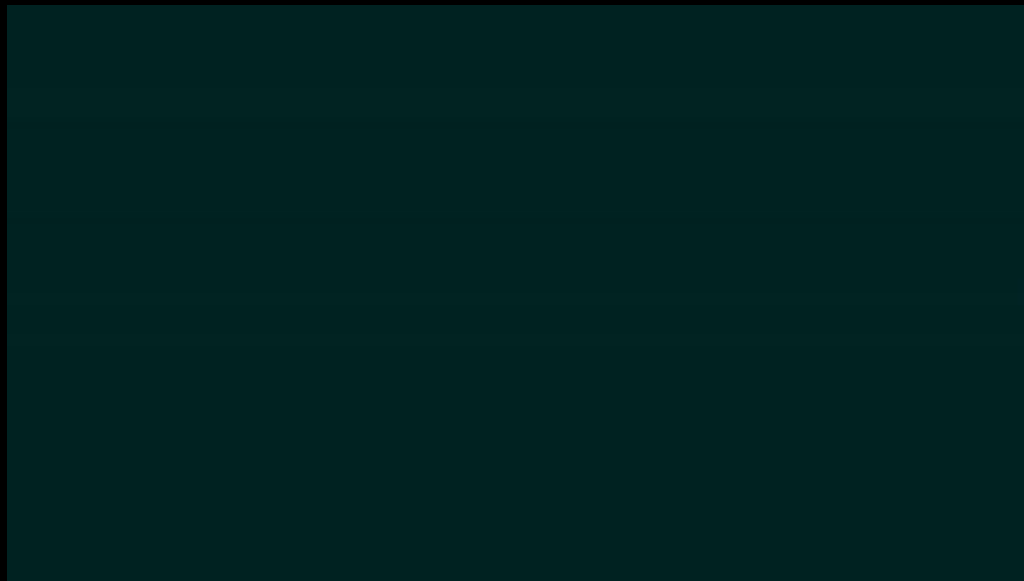
# GTO Wizard AI

| Agent                               | Slumbot (2018)                   | Top Humans     |
|-------------------------------------|----------------------------------|----------------|
| DeepStack's reimplementation (2020) | $-6.3 \pm 4.0$                   |                |
| ReBeL (2020)                        | $4.5 \pm 0.5$                    | $16.5 \pm 6.9$ |
| Supremus (2020)                     | <b><math>17.6 \pm 4.4</math></b> |                |
| GTO Wizard AI (2022)                | <b><math>19.4 \pm 4.4</math></b> |                |

Table 2: Head-to-head results showing expected winnings (bb/100).  
The  $\pm$  shows one standard deviation.



# GTO Wizard AI



Results vs. Slumbot across 150k hands

# GTO Wizard AI – Public API

Over the last few years, there has been a lack of benchmarks and transparency in the field.

→ Which is why we have decided to release a public API where you can now play against our solver, completely free of charge.

We're currently testing it for heads-up, but we will eventually also allow different formats to be played like multiplayer Hold'em and PLO.



# Heads-up Tournament

As a way to support the MIT pokerbots competition and to celebrate the beta release of our API, we launched a side quest for you:

Compete against our heads-up NLH AI and win prizes!

Here are the prize pool and the details of the challenge:

- 💰 Prize Pool: 1st: \$2500 | 2nd: \$1500 | 3rd: \$1000
- 📋 Requirements: Play 3000 hands before January 30th against GTO Wizard AI
- No cheating is allowed (no use of external charts or outputs from other solvers)



# Heads-up Tournament

- Note that results will be measured by the AIVAT metric (Burch & al., 2017)
- AIVAT is an unbiased variance reduction technique developed by researchers at the University of Alberta
- AIVAT achieves a 3x reduction in standard deviation, which allows us to achieve the same statistical significance with ten times less data.
- As such, while we require only 3k hands to be played



# Heads-up Tournament

Can your bot beat top LLMs?

| Model                  | Winrate (bb/100) |
|------------------------|------------------|
| Always Fold            | -62              |
| GPT-4                  | -96              |
| GPT-4o                 | -39              |
| GPT 5.2 High reasoning | -15              |
| Gemini 3 Pro Preview   | -23              |



# What's next?

**Human modeling:** Trying to approximate a Nash Equilibrium in multiplayer and non zero-sum settings doesn't guarantee you to win in general. Incorporating human modeling into the solver and learning how to exploit players is an essential part of multiplayer poker.

**Scale to larger games:** ReBeL-like algorithms don't scale well to games with very large beliefs size, which is common:

Omaha (PLO): 270,725 or 2,598,960 possible hands

Drawing games like Seven Card Stud, where you can draw up to 7 cards, for a total of 133,784,560 possible hands!

→ Solve any poker game to a high accuracy in a few seconds

# Thank you!

There's still a lot of exciting research and development to be done in the field.

If being at the forefront of technology sounds interesting to you, please reach out to me at [phil@gtowizard.com](mailto:phil@gtowizard.com)



**GTO Wizard**

# Lunch 🤪

Leave any type of feedback at [pkr.bot/feedback!](https://pkr.bot/feedback)





[pkr.bot/gto](#) (GTO Wizard challenge)

[pkr.bot/hack](#) (2nd Hackathon interest)



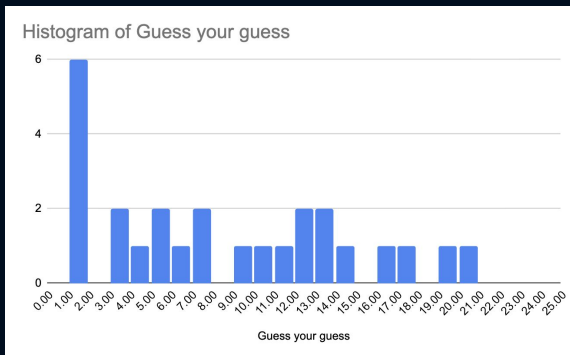
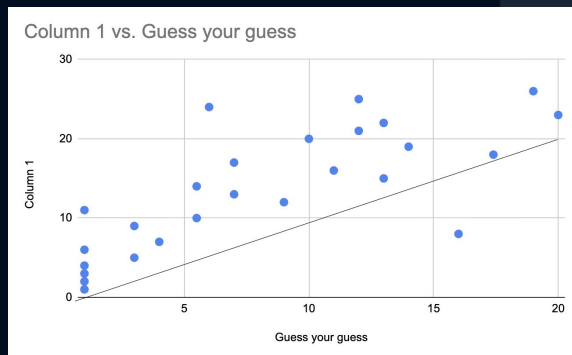
Giveaway  
Winner

# Guess Your Guess Winner:

*ghaisman*

Some cool facts:

- 2 people guessed before the link was up C:
- 6 people guessed '1'
- Nic76 guessed 17.4, was 18th (2nd place)



# Thanks for watching!

Information hub at [pkr.bot/hub](https://pkr.bot/hub)

Slides/notes will be posted on [pkr.bot/resources](https://pkr.bot/resources)

Make sure to check [pkr.bot/piazza](https://pkr.bot/piazza) for updates

Lecture recordings at [pkr.bot/panopto](https://pkr.bot/panopto)

Leave feedback at [pkr.bot/feedback](https://pkr.bot/feedback)!